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Extracting Triples from Dialogues for Conversational Social Agents

Piek VOSSEN [a1], Selene BÁEZ SANTAMARÍA [a] Lenka BAJČETIĆ [b] and Thomas BELLUCI [a]

[a] *CLTL, Vrije Universiteit Amsterdam*
[b] *Inovacioni Centar ETF: Belgrade, RS*
ORCiD ID: Piek Vossen https://orcid.org/0000-0002-8273-0132, Selene Báez Santamaría https://orcid.org/0000-0003-3808-2614, Lenka Bajčetić https://orcid.org/0000-0002-2005-5426

Abstract. Understanding of communication is essential to create controllable and transparent agents for Hybrid Intelligence collaboration. In this paper, we describe a number of Natural Language Understanding models that extract explicit symbolic triples from social conversation. Triple extraction has mostly been developed for Knowledge Base Completion using Wikipedia data for training and testing. However, social conversation is a very different genre. Phenomena such as co-reference, ellipsis, coordination, and implicit and explicit negation or confirmation are more prominent in conversation. We therefore fill this gap by releasing data sets for triple extraction from social conversation. We created five new triple extraction models and tested seven models on our data. The highest precision was achieved by a generative model, Qwen2.5: 54.02 for complete triples and 73.18 for triple elements tested on single utterances. However, scores for conversational triples that span multiple turns are significantly lower, showing that extracting knowledge from true conversational data is challenging.

Keywords. Information extraction, Social Conversations, Conversational AI

1. Introduction

In a Hybrid Intelligence context in which humans and agents collaborate, social information exchange is crucial. Recently, generative Large Language Models (LLMs) [1,2] excel at having social conversations; yet the knowledge exchanged remains implicit. Our goal is to develop conversational agents that explicitly represent the information exchanged in a way that can function as an episodic memory, for example, as Knowledge Graphs (KGs) [3]. This will give an agent more control over the goals of the interaction and allow for explicit symbolic reasoning. Especially in collaborative scenarios, such an explicit representation can provide transparency and explainability. In order for social conversational agents to create such KGs, they need to detect information and perspectives expressed in dialogue as RDF[2] triples that can be added to the memory graph.

[2]Resource Description Framework: https://www.w3.org/RDF/

People often talk about themselves in social conversations. Most conversations are therefore homodiegetic [4], telling a story from the perspective of the speaker. It is not surprising that the most frequent subjects and objects in conversational utterances are the pronouns "I" and "you" referring to the speakers [5]. However, little is known about the information that is exchanged during such conversations. Hence, we also do not know what conversational social agents need to understand from such conversations.

Most of the state-of-the-art technology for extracting explicit knowledge is concentrated on Wikipedia texts, primarily because of resource availability. However, social conversation contains a mixture of facts and personal perspectives [5] that are expressed through sequential dialogue turns between interlocutors. As partial information is provided by different speakers, it is important to deal with coreference across these utterances, ellipsis, coordination, implicit and explicit negation, and speaker perspectives such as certainty of beliefs, judgments and emotions as values behind beliefs. This makes it challenging to develop conversational agents that can have social conversations with humans and extract the explicit knowledge provided.

In this work, we focus on the knowledge exchanged in social conversation. We define training and test data for the task of triple extraction. We compare several models, ranging from dedicated context-free-grammars, dependency parsing, to fine-tuned and prompted LLMs[3]. Our contributions are:

- An English test suite for dialogue turns annotated with social facts and perspectives as triples, categorized by dialogue acts and types of knowledge.
- Training data for dialogue turn sequences extracted from PersonaChat, DailyDialog, and Circa annotated with factoid triples and speaker perspectives.
- Five new models for the triple extraction of social conversation and test results for seven models.

2. Related work

Information extraction is the task of identifying factual information in unstructured data and representing this knowledge in a structured form. Knowledge graphs (KGs) are one of such forms, where nodes represent entities and edges represent relationships between these entities. In that framework, a fact is then represented as a triple: *subject - predicate - object* [6].

Types of information extraction Information extraction can be divided into two categories: Closed Information Extraction (CIE) and Open Information Extraction (OIE). CIE aligns the extracted information with existing entities, relations, or attributes in a predefined KG. This approach relies on (semi-)supervised methods to extract triples, often combining Named Entity Recognition, Entity Linking, and Relation Extraction. CIE systems are typically tailored to a specific domain and KG schema, such as SKOS or DBpedia [7]. Therefore, the KG schema determines the set of recognizable entities and relations a priori. In contrast, OIE is not limited to a predefined set of entities and relations. Instead, it involves extracting spans from text that represent entities and their

[3] All our data and models are freely available on GitHub: https://github.com/leolani/cltl-knowledgeextraction, https://github.com/leolani/cltl-conversational-triple-extractor

relationships without prior alignment to a KG. This open nature makes OIE suitable for extracting new or domain-independent knowledge.

Historical perspective Information extraction is a well-established research area that has evolved significantly over time. The early work was manual-intensive and rule-based methods were developed primarily out of linguistic insight[8]. Most of these rule-based systems focused on extracting is-a relations, thus forming taxonomies in specialized domains. Over time, the field has shifted to feature-based systems, with the emergence of OIE. Developing robust general-purpose relation extractors continues to be an open challenge in the field [9]. Most recent approaches use generative models and prompting to extract information from text. Zhang et al. [10] introduce the Generative IE framework, in which structured information is generated from text as a decoding task. They argue that the same framework can be used for various IE tasks, including complex structures, and has shown competitive performance. The downsides are excessive compute and low speed. Furthermore, such models have not been tested on social conversational data that we are dealing with. Despite advances, relation extraction remains a significant bottleneck, often yielding lower precision compared to Named Entity Recognition.

Extracting personal knowledge The task of DialogNLI was introduced by Welleck et al. [11] to determine whether pairs of utterances from social conversations entail, contradict or are unrelated to one another. For this purpose, they annotated single sentences from the PersonaChat dataset [12] with triples and possibly negation values, e.g., (I, like_drink, espresso). However, Welleck et al. did not train a triple extractor from these data, but used the triple annotations to train a classifier to infer the NLI relationship between turns represented through these triples. Wang et al. [13], on the other hand, did use the annotated DialogNLI data to fine-tune GPT-2 to extract structured information about a person, such as their hobbies, pets, family, likes and dislikes, which we define here as **social facts**, in the form of triples. Their approach comes closest to our work, but whereas they consider single turns to extract implicit properties, we consider both single turns and conversational sequences of turns that are more complex. Furthermore, they propose a single generative model, and we experimented with various smaller models and produced a new data set both for turn-level and conversation-level extraction.

3. Conversational information exchange

The extraction of world knowledge and personal information from dialogue presents unique challenges from a linguistic point of view. Whereas alternative domains, such as news and Wikipedia articles, can be presumed to consist of complete grammatical sentences, dialogues are often highly fragmented and can give rise to many kinds of ungrammatical non-sentential constructions [14,15]. Dialogue is defined as a joint activity involving two or more participants in which information or ideas are exchanged through natural spoken or written language [16,14]. In this view, the primary purpose of dialogue is simply to provide a means to share information, build common ground, and allow one to learn about others, their beliefs, and the world around them. In particular, the perspectives of the interlocutors on social facts as defined in the GRaSP model [17,18,19] add another layer of complexity, both in terms of language structure and status of information. Interlocutors can express their epistemic beliefs (confirm or denial), their feelings and emotions, and the certainty of any belief using more complex linguistic constructs.

In the following examples, taken from the DailyDialog dataset, we highlight several noteworthy dialogue phenomena including ellipsis, the use of anaphora, and the speaker's perspective.

Explicit Negation In the next example, "Speaker 2" asks a direct question to"Speaker 1" and receives a negative answer (polarity -1) with a negative sentiment (-1), showing different perspective values for the claim of *having kids*. We see here that social facts and perspectives are mixed and need to be inferred from multiple turns across interlocutors.

Example 1: Explicit negation represented as RDF triples

Speaker 1: Ow wow now that will be hard task
Speaker 2: Yeah do you have kids?
Speaker 1: No, I sadly do not. I never been one to hold a steady relationship

triple: (speaker1, have, kids)
perspective: polarity=-1, certainty=01, sentiment=-1

Implicit Negation The examples below are more complex, showing two dialogues with implicit negation. In the first,"Speaker 1" states that they do not have time to watch a new drama, which only indirectly negates watching it (polarity -1) with high, not absolute, certainty (0.8). The sentiment is still positive (1) as we can assume that"Speaker 1" would like to see it. In the second fragment,"Speaker 1" does not directly answer the question of"Speaker 2" but makes another statement that makes it unlikely that"Speaker 1" likes living there. In this case the polarity is also negative, but certainty is expected to be lower, and, in general,"Speaker 1" expresses negative sentiment towards living there.

Example 2: Implicit negation represented as RDF triples

Speaker 1: I do work! I waitress during the day, then I do side jobs during the evening.
Speaker 2: Will you be watching that new drama this evening?
Speaker 1: I do not have any time to start a new series today.

triple: (speaker1, have, kids)
perspective: polarity=-1, certainty=0.8, sentiment=1

Example 3: Implicit negation with coreference represented as RDF triples

Speaker 1: Yup been to new york city 3 times this year.
Speaker 2: Do you want to live in New York city?
Speaker 1: The city is too busy and loud.

triple: (speaker1, like, living_in_NY)
perspective: polarity=-1, certainty=0.5, sentiment=-1

Triple elements are spread over the turns, such as "watching" and "like living_in_NY" in turn 2 of both conversations. We also observe coreference of "the city" with "New York city" and coordination of two claims about NY, being both too busy and too loud.

To capture phenomena such as the above, we created two data sets with conversational utterances. The social facts expressed are annotated as triples, along with the per-

spective information of the sources. We first describe a test suite with single turns and, second, annotations of existing conversational data between crowdworkers annotated in a similar way over sequences of turns.

3.1. Test suite of turns

We built a dedicated dataset with single-turn statements and questions given human and agent speaker roles. The data were specifically designed to capture distinct domains and linguistic phenomena. For example, we created datasets for open verb questions and wh-questions separately. This diverse data tests the capabilities of each extractor on a wider range of conversational topics. Each test item consists of the text and a representation of the triple as in the following examples, where the human speaker is called ”Lenka”:

- I have three white cats: lenka have three-white-cats
- can I make a cake: lenka make a-cake
- who is from Mexico: ? be-from Mexico
- what do you enjoy: agent enjoy ?

Example 1 is a statement, which entails an RDF statement to be added to the KG. Example 2 is a verb-question, that should be posted as a SPARQL query to retrieve a yes/no answer. Examples 3 and 4 are wh-questions with one variable in the triple, also to be posted as SPARQL queries resulting in a (possibly empty) list of answers.

In total, 87 statements and 48 verb-questions are created with implied triples, and 58 wh-questions with triples that contain one variable. Finally, we created 27 statements in which also perspectives are expressed in relation to social facts. In that case, we add the perspective values for certainty, polarity, and sentiment as scored values in the respective order, as shown below.

- john doesn’t like fashion: john like fashion: 1.0 -1.0 -1.0
- I think that birds can fly: birds like flying: 0.75 1 0.0
- I hope bob is from england: bob be-from england 0.5 1 1.0

3.2. Test suite of conversations

Most triple extraction solutions only consider single sentences or, in the case of conversation, the information expressed in a single turn. However, as explained above, conversations are often more complex than written text such as Wikipedia or news, while information is also exchanged over multiple turns. We therefore annotated a dataset with triples that span sequences of three turns. We sampled data from three complementary sources: PersonaChat [12], DailyDialog [20] and Circa [21].

We randomly selected 941 short dialogue fragments of three consecutive utterances: 442 dialogues from Circa, 374 dialogues from DailyDialog and 301 from PersonaChat. The utterances approximately have a length of 29 tokens, excluding end-of-turn markers. Three annotators (authors) annotated the dialogues for social facts by marking the subject, predicate, and object tokens, representing triples claimed by one of the interlocutors. In addition, the annotators could mark any tokens expressing the polarity and certainty of the interlocutor making the claim. Sentiment was not annotated in this data set. Table 1 shows an overview of the data and annotations.

Table 1. Overview of the conversational dataset with triple annotations

Total Dialogues	1,117	Annotators	3
Turns per Dialogue	3	Total Triples	4,786
Total Utterances	3,351		
Total Tokens	32,877		
Tokens per Dialogue	29.43	Triples per Dialogue	4.28
Tokens per Utterance	9.81	Triples per Utterance	1.38

The 941 dialogues contained 2,823 utterances, which received a total of 3,899 annotations of triples and perspectives. We calculated the Inter-Annotator Agreement on a pilot set in terms of the Jaccard index and average F measure across the annotators, averaging over all annotator pairs and dialogues. Annotation agreement was measured at two levels; the level of triple elements (that is, the sets of subjects, predicates, and objects marked in a dialogue) and the level of triples. The agreement was Jaccard=0.688 and F-measure=0.773 on the identification of whole triples. When analyzing the triple elements separately, there is a slight drop in performance on predicates and objects.

The annotated data set is divided into training and test sets with a 9:1 ratio per dialogue, to ensure that overlapping fragments of the same dialogue fall within the same set. We manually selected 179 dialogues (537 utterances) for testing, exhibiting different linguistic phenomena that characterize conversational information exchange: answer ellipsis, coordination, coreference, statements, negated statements, explicit no-answers and explicit yes-answers. Each of these constitutes a separate test set. A few examples are shown next, where three turns are separated by ”EOS” tokens in the order of ”speaker1”, ”speaker2” and ”speaker1”. The triple is given between brackets followed by a perspective label for polarity:

- **dailydialogs-train-012081**: Yes , it’s me. EOS Do you have a cold ? EOS No. Worse than that. I have a flu. I’m in bed with a fever.(speaker1, have, a cold, polarity-negative)
- **circa-train-002436:**: alot really, i enjoy going out to eat with family, going to the movies EOS would you like to go to the movies tonight ? EOS no, i’m busy (speaker1, like to, go to the movies, polarity-negative)
- **personachat-valid-000114**: ow wow now that will be hard task EOS yeah do you have kids? EOS no i sadly do not. i never been one to hold a steady relationship (speaker1, have, kids, polarity-negative)
- **circa-train-010351:** we could put on some christian edm and party once your taller to celebrate EOS do you like edm ? EOS no, i don’t know them (speaker1, like, edm, polarity-negative)

4. Triple extraction models

We created different models to extract both factual information and perspectives expressed in dialogues. In particular, we tested five types of triple extraction approaches, among which existing models that were not developed for this type of data.

1. Tailored Context-Free Grammar (CFG)
2. spaCy Dependency Parser

3. Stanford OpenIE tool
4. Two fine-tuned BERT-based models
5. Two generative Large Language Models (Llama and Qwen) prompted with few shot learning

Each of these model types utilizes different NLP techniques to extract subject-predicate-object triples and, possibly, their associated perspective values. In addition, they differ in the language genre for which they were built. Encoder models such as BERT and ALBERT have been pre-trained with written texts such as Wikipedia, news and books and not with conversational data. Also, spaCy and OpenIE have been built to process written text instead of conversation and have not been adapted specifically to this task. BERT and ALBERT have been fine-tuned with conversational data, whereas generative models such as Llama and Qwen have been pre-trained with a large variety of language, including conversational data. Comparing these models will tell us more about the design factors that matter for performance. The Context Free Grammar is used as a baseline as it was specifically built to deal with the type of sentences in the single-turn test data.

4.1. Dependency parser

We used the spaCy dependency parser and simple dependency patterns mapped to semantic relations as a baseline system for a system that is not adapted. Patterns of frequent phrase structures with these dependency relations are converted into subject-predicate-object triples [13]. We extract triples for active and passive variants of subject-predicate-object ("cats are liked by John"), subject-predicate-adjectival-complement ("John is sick"), subject-predicate-prepositional-object ("John goes to school").

4.2. Context Free Grammar

We developed a dedicated Context Free Grammar (CFG) specifically targeting phrase structures of simple statements and questions as single turns. The CFG-based extractor is a rule-based system designed to process individual utterances and identify subject-predicate-object triples. It uses a pipeline of tokenization, Part-of-Speech tagging using both NLTK and Stanford taggers, and Named Entity Recognition(Stanford NLP). The processed tokens are passed through a manually designed context-free grammar, which parses the sentence structure to identify triples. The grammar has 12 rewriting rules to define basic phrase structures, which can be found in our Github repository.

The output of the CFG parsing is mapped to subject-predicate-object triples following predefined rules. Tokens are further lemmatized using NLTK, and modal verbs are analyzed with a lexicon to capture additional perspective values, such as uncertainty or temporality. Multiword expressions, such as New York or ice-cream, are detected as collocations and treated as single entities.

Finally, from the phrase structure, we extract triples using specific patterns and a lexicon with 1) lexical categories such as pronouns, (auxiliary) verbs, determiners, quantifiers, 2) semantic categories such as activities, professions, and kinship terms, and 3) sentiment. This detailed semantic analysis ensures that each triple is enriched with meaningful contextual information.

4.3. Conversational analyzer

We fine-tuned two BERT models (multilingual BERT-based and Albert) with the annotated sequences of three alternating turns between interlocutors described in Section 3. These models are specifically trained for IOB-style triple extraction and are adapted for conversational contexts by incorporating sequences of three turns: speaker-agent-speaker. The IOB extractor generates all candidates for subjects, predicates, and objects from the three utterances and ranks these using a trained ranker. This approach is particularly suited to conversational settings, capturing the nuances of how people exchange knowledge. The training data and the settings for fine-tuning are available on our Github.

4.4. Few-shot LLM prompting for triple extraction.

We also prompted two generative models, Llama3.2 (3B parameters) and Qwen2.5 (7B parameters), to extract both triples and perspectives. We created separate prompts for statements and questions. We gave examples for statements (4), WH-questions (5) and a verb-question (1) from our turn-level test set for few shot learning. A separate prompt was created for triple extraction of conversations, taking three turns as input, similar to the conversational analyzer. The prompts can be found in the Github repository.

4.5. Stanford Open Information Extraction

For comparison, we also tested a state-of-the-art triple extractor that was trained on Wikipedia data on our test data. The Stanford OpenIE extractor is designed to extract general relationships from text. Unlike traditional IE tools that rely on a broad set of patterns, this approach simplifies the process by using a smaller set of patterns tailored for canonically structured sentences [22] .

To handle longer and more complex sentences, the extractor employs a classifier that isolates self-contained clauses. These clauses are then processed using natural logic inference to identify the most specific arguments for each candidate triple. This approach ensures that the extracted triples are both precise and contextually relevant.

Although the model is trained on Wikipedia, and thus is encyclopedic, it tends to produce a high volume of triples, including noisy or irrelevant ones. Common issues include the extraction of unnecessary modifiers or articles. Post-processing is applied to refine the triples and filter out such noise, improving the overall quality of the output.

5. Results

In this section, we present the result on our turn-level and conversation-level test set. Only our conversational models could be applied to the latter. Adapting the Llama prompt for conversational contexts is left for future work.

5.1. Turn level

Table 2 shows the model results in the turn level test, where we calculate the precision by taking the number of correct triples divided by the sum of correct and incorrect triples. We calculate the precision for the complete triples, the elements of the triples regardless

of their combination, and the precision for each element: subject, object, and predicate. In most cases, we see that the models perform consistently on the complete triples, the triple elements, and each of the triple components. An exception to this is formed by Llama and openIE for verb-questions. For the former, this is due to the lower performance on subjects, and for the latter to the low performance on predicates.

The Qwen model outperforms all other models significantly, for statements, verb-questions and wh-questions. Note that this performance of Qwen2.5 comes with a price: the model requires a lot of memory (4GB) and is very slow compared to the encoder models, which is problematic for a conversational agent. The CFG, convBERT and Llama models perform more or less similarly for statements as second. Llama performs a bit better on objects and predicates but lower on subjects than CFG. Llama is prompted in the same way as Qwen so the model (3B versus 7B parameters respectively) and its pretraining make the difference here. The convALBERT scores significantly lower than convBERT even though both models are trained the same way with the same data. Again, the model and pre-training make the difference. Possibly, the fact that convBERT is a **multilingual** model makes it more robust for other genres, such as dialogues, giving it an advantage over monolingual models (ALBERT). For questions, the difference between convBERT and convALBERT disappears. The CFG model performs second for wh-questions, probably due to the fact that a wh-structure is relatively easy to capture through grammar rules. The openIE and spaCy models perform very poorly compared to the others in all three test sets.

We carried out an error analysis on the best performing model types: Qwen, CFG and convBERT. For Qwen, the most frequent errors were empty subjects (14), objects (13), and predicates (17), making triples incomplete. Other difficulties for the model are to interpret the complementation of the verb ”be” and to resolve friendship and kinship relations relative to the speaker (15). E.g. in the case of ”Carl is the brother of Carla” the predicted predicate is ”be” but should be ”brother-of” and in the case of ”Carl is my brother”, the predicate is ”brother-of” but the object is the speaker. The convBERT model had less empty slots (1 object and 9 predicates). Most errors originate from including a determiner in the object (”a-banana” instead of ”banana”), resolving friendship and kinship relations relative to the speaker (38) and including epistemic and deontic auxiliaries (”think”, ”should”, ”must”, ”might”) in the predicate (12). The CFG also suffers a bit from empty subjects (1), objects (3), and predicates (8). Other errors are determiners included in objects (10), friendhip/kinship (85), ”be” complementation (29) and auxiliaries included in the predicate (46). Many of the errors are related to the normalization of the triple elements, both objects and predicates. In future work, we want to extend the triple extraction to include IRIs from the KG instead of relying on the label representation. This requires matching the text labels to labels already in the Knowledge Graph, which allows learning to link and correcting mappings in an interactive conversation using user feedback, e.g. ”Is a banana the same as banana?”.

Some of the models (CFG, the conversational models and the LLMs) can also extract the perspective on the statement expressed by the speaker, such as polarity, certainty, and sentiment. We therefore created a test set of 27 utterances in which the speaker not only expresses a statement but also a social fact, indicating their epistemic and emotional belief, e.g. ”I think he does not like sports”. The results are shown in Table 3. Statements with perspectives have precision scores for both the triples and the perspectives.

Table 2. Precision of the triple extraction on the utterance level test for seven models: CFG=Context Free Grammar, convBERT= multilingual BERT-base fine-tuned with conversational triples, convALBERT=Albert-base finetuned with conversations, Llama3.2 prompted with few-shots, Qwen2.5 prompted with few-shots, openIE=Stanford's open information extraction, spaCy=spaCy dependency patterns. Three different test sets are given for statements, verb-questions and wh-questions. Columns present the precision of extracting the complete triples, any of the triple elements regardless of the combinations, the subjects, the objects and the predicates.

test	# utt.	model	Triple	Triple-element	Subject.	Object	Predicate
statements	87	CFG	25.29	54.41	73.56	57.47	32.18
		convALBERT	12.64	36.40	44.83	39.08	25.29
		convBERT	17.24	57.09	75.86	58.62	36.78
		Llama3.2	26.44	54.79	42.53	65.52	56.32
		Qwen2.5	**54.02**	**73.18**	**79.31**	**68.97**	**71.26**
		openIE	2.30	34.48	47.13	45.98	10.34
		spaCy	31.03	44.06	42.53	42.53	47.13
verb questions	48	CFG	16.67	57.64	79.17	56.25	37.50
		convALBERT	18.75	47.22	54.17	35.42	52.08
		convBERT	18.75	53.47	62.50	45.83	52.08
		Llama3.2	6.25	43.75	10.42	45.83	75.00
		Qwen2.5	**45.83**	**75.00**	**81.25**	**58.33**	**85.42**
		openIE	0.00	29.17	54.17	33.33	0.00
		spaCy	29.17	47.92	52.08	33.33	58.33
wh questions	58	CFG	29.31	58.05	60.34	82.76	31.03
		convALBERT	15.52	48.85	56.90	60.34	29.31
		convBERT	20.69	41.95	41.38	62.07	22.41
		Llama3.2	18.97	45.40	29.31	48.28	58.62
		Qwen2.5	**56.90**	**74.71**	**72.41**	**81.03**	**70.69**
		openIE	0.00	3.45	8.62	1.72	0.00
		spaCy	1.72	2.30	3.45	1.72	1.72

Table 3. Precision of the triple extraction on 27 utterances expressing various perspectives (certainty, denial, sentiment) of speakers on the statements in utterances. Results for five models: CFG=Context Free Grammar, convBERT= multilingual BERT-base fine-tuned with conversational triples, convALBERT=Albert-base fine-tuned with conversations, Llama3.2 and Qwen2.5 prompted with few-shots. Columns present the precision of extracting the triples and the perspective as the complete tuples of values and separately for certainty, polarity and sentiment.

model	Triple	Triple elem.	Subject	Object	Predicate	Perspective	Certainty	Polarity	Sentiment
CFG	25.93	44.44	55.56	51.85	25.93	**66.67**	71.43	80.95	**47.62**
convALBERT	7.41	30.86	37.04	37.04	18.52	60.00	**100.00**	40.00	40.00
convBERT	18.52	53.09	62.96	66.67	29.63	52.94	76.47	47.06	35.29
Llama3.2	40.74	59.26	55.56	66.67	55.56	46.03	23.81	71.43	42.86
Qwen2.5	**48.15**	**77.78**	**81.48**	**74.07**	**77.78**	55.56	23.81	**100.00**	42.86

We expect the extraction of the triples to be more difficult because the linguistic structure is more complex and reflects different layers of information. For all models, we see that the results for the triples are, in fact, a bit lower compared to the results for the statements in Table 2. However, for generative LLMs, this difference is not as consistent, where Llama performs better in extracting subjects and complete triples. If we look at the

actual perspective values, we see that the generative LLMs no longer always outperform the other models. Qwen outperforms the others only for polarity.
We conducted an error analysis on the predicted perspective values. The first observation is that generative LLMs generate a much larger variety of values. For example, even though we explicitly instructed that polarity can only have the values 1.0 (claim) and -1.0 (denial), Llama still generates values such as 0.0 and 0.5 for polarity. The values for certainty and sentiment are also more nuanced. The latter cannot always be considered as real errors since some scalar judgments (0.9 != 1.0) are subjective. Other models are programmed to generate only very basic values, e.g. -1 (negative), 0 (neutral), and 1.0 (positive) for sentiment. Since the annotations are also very basic, this gives these models an advantage. In future work, more nuanced annotations should be tested.

5.2. Conversation level

The conversational models were designed to extract triples from contextual sequences of turns. Table 4 shows the results for the best models convBERT and Qwen2.5, where the test data are differentiated for separate linguistic phenomena.

Table 4. Precision of the conversation level evaluation for the conversational model. The test is divided over different conversational phenomena such as ellipsis, coordination, coreference, statements, negations of statements, no answers and yes answers.

test	utt.	model	Triple	Triple elem.	Subject	Object	Predicate
answer ellipsis	101	convBERT	0.99	21.78	52.48	5.94	6.93
		Qwen2.5	**2.97**	**26.40**	**60.40**	**8.91**	**9.90**
coordination	82	convBERT	**6.10**	**32.93**	58.54	**21.95**	18.29
		Qwen2.5	2.44	31.71	**63.41**	8.54	**23.17**
coreference	78	convBERT	**1.28**	**17.09**	26.92	**20.51**	3.85
		Qwen2.5	0.00	14.96	**28.21**	10.26	**6.41**
statements	91	convBERT	**1.10**	**28.21**	**58.24**	**18.68**	7.69
		Qwen2.5	**1.10**	25.64	**58.24**	8.79	**9.89**
statements	79	convBERT	1.27	**27.85**	54.43	**20.25**	**8.86**
negated		Qwen2.5	**2.53**	24.47	**55.70**	8.86	**8.86**
explicit	47	convBERT	2.13	23.40	51.06	**8.51**	10.64
no answers		Qwen2.5	**4.26**	**27.66**	**59.57**	**8.51**	**14.89**
explicit	59	convBERT	**1.69**	23.73	54.24	**5.08**	**11.86**
yes answers		Qwen2.5	0.00	**24.29**	**62.71**	1.69	8.47

Overall, the results are lower than for the turn-level evaluation, which is expected, as this is a more challenging task. The precision at the complete triple level is very low and the precision for the predicates is the lowest from the elements. The subject elements score highest in most cases, while for conversations exhibiting coreference, subject scores are also lower. Most of these subjects are expected to refer to the interlocutors, which are more easily detectable by mapping the pronouns "I" and "you" correctly. The two models do not consistently outperform each other. There is a slight gain by Qwen but in various cases convBERT outperforms Qwen.

We carried out an error analysis on all mismatches. We found the following errors for subjects and objects: normalization errors on how to represent labels (14-yuan != 14 yuan), true differences (apple != dolls), related through synonymy, hyponymy

or metonymy (band != heavy metal), possessive constructions involving the speaker (speaker1 != speaker1 's sisters), unresolved coreference (it != speaker1 's hat, : he != justin). For predicates, we also found errors due to auxiliaries (be != paying with), negations (am-not != am) and quotes ('-s != are).

Table 5. Error analysis for triple elements. Proportion of errors per error type. Lower proportions are better.

element	model	normalization	different	related	possessive	coref	aux	negation	other
subject	convBERT	10.57%	52.85%	1.63%	4.07%	30.89%	n.a.	n.a.	n.a.
	Qwen2.5	3.57%	25.00%	1.19%	20.24%	50.00%	n.a.	n.a.	n.a.
object	convBERT	20.87%	63.88%	13.79%	1.45%	0.00%	n.a.	n.a.	n.a.
	Qwen2.5	10.67%	68.00%	17.33%	0.00%	0.00%	n.a.	n.a.	n.a.
predicate	convBERT	14.29%	66.85%	7.82%	n.a.	n.a.	5.39%	0.54%	5.12%
	Qwen2.5	13.56%	59.32%	5.08%	n.a.	n.a.	4.24%	10.17%	7.63%

In terms of errors, we see considerable differences between models and between the triple elements. The convBERT model suffers more from normalization errors and true differences in subjects, whereas Qwen suffers more from interpreting possessive constructions, co-references and negation for predicates. Further fine-tuning or prompt engineering could address these errors. The true differences mean that the models picked up very different elements from the utterances. This is not necessarily wrong since each model extracts multiple triples but selects only one. It could very well be that the right triple is still extracted but not prioritized.

6. Conclusion

Triple extraction from social conversations is more difficult than extraction of information from news or Wikipedia texts. First, information is spread over different turns across interlocutors, through incomplete sentences that exhibit ellipsis, coreference, coordination, and implicit statements. Second, the information itself is more complex in social exchange compared to news and Wikipedia facts. It involves a wider and more open range of properties and values that address social relationships, personal feelings, and history.

We released two data sets for training and testing the triple extraction of social conversation. We also developed several models, among which two models for the difficult task of extracting social facts and perspectives from ongoing conversations involving sequences of turns. We hope that this enables future research to model social conversations more explicitly. So far we tested the models only on English. The Llama, Qwen and multilingual convBERT model could be applied to data in other languages. In future work, we plan to extend the evaluation and training data to other languages than English.

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Appendix

Turn-level evaluation examples

- Statements
  - I know you: lenka know agent
  - where is she: she be ?
  - my best friend is he: lenka best-friend-is he
  - I have three white cats: lenka have three-white-cats
  - I think Selene doesn't like cheese: selene like cheese
  - I think Selene hates cheese: selene hate cheese
  - selene might come today: selene might-come today
  - I don't think selene likes cheese: selene like cheese
- Verb-questions:
  - can a bird sing a song: a-bird can-sing a-song
  - can I call you: Lenka can-call agent
  - can I make a cake: Lenka can-make a-cake
  - will you go to Paris: agent will-go-to paris
  - must you go home: agent must-go home
  - do you like amsterdam: agent like amsterdam
- Wh-Questions
  - who is your best friend: agent best-friend-is ?
  - who can sing: ? can-sing ?
  - who likes talking to people: ? like talking-to-people
  - who works at the university: ? work-at the-university
  - who have you seen: agent see ?
  - who is from Mexico: ? be-from Mexico
  - who does Selene know: Selene know ?
  - what is your favorite color: agent favorite-color-is ?
  - what is my favorite food: Lenka favorite-food-is ?
  - what do you enjoy: agent enjoy ?
  - where is selene from: Selene be-from ?

  * where is your friend: agent friend-is ?

- Perspectives
  * john doesn't hate fashion: john hate fashion: 1 -1 -1
  * I think selene works in Amsterdam: selene work-in amsterdam: 0.75 1 0
  * I think john can't come to school: john can-come-to school: 0.75 -1 0
  * you know I like coffee: lenka like coffee: 1 1 0.75
  * I think that birds like flying: birds like flying: 0.75 1 0.75
  * I know that you are not a human: agent be a-human: 1 -1 0
  * you must bring three books: agent must-bring three-books 1 1 0
  * john might like reading books: john might-like reading-books 0.5 1. 0.75
  * selene should come to the university: selene should-come-to the-university 0.75 1 0
  * Bob might be from england: bob might-be-from england 0.5 1 0

Context-Free Grammar rules

```
U -> S | Q
Q -> W VP C| W NP C | W VP NP C | W VP NP VP C | W NP VP C | V NP C | V NP C C
W -> WRB | WP | WDT
S -> NP VP C | NP C | NP VP S
C -> NP | VP | PREP
NP -> N | JJ | J NP | D NP | N NP | RB NP | PREP NP
VP -> V VP | V RB | RB VP | V | V PREP VP | V PREP
PREP -> IN | TO
V -> VBD | VBP | VBZ | VBN | VBG | VB | MD
D -> DT | CD | PRPPOS
N -> NN | NNS | NNP | NNPS | PRP | DT
J -> JJ | JJR | JJS
```

Prompts with few shot examples

```
class STATEMENT:
    INSTRUCT = {'role':'system', 'content': '''You will receive a statement from a user in a conversation
and you need to break it down into triples consisting of a subject, predicate and object.
Each triple should capture the essence of statement by the speaker.
Replace the predicate by its lemma, for example "is" and "am" should become "be", "likes" and "liked" should become "like".
Remove auxiliary verbs from the predicates such as "be", "have", "can", "might", "must", "will", "shall", "should",
and also negation variants such as "cannot", "won't", "shouldn't".
Remove negation words such as "n't", "no", "not" and "never" from the predicates.
If the object starts with a preposition, concatenate the preposition to the predicate separated by a hyphen,
for example "I am from amsterdam" should become {"subject": "I", "predicate": "be-from", "object": "amsterdam"}.
If "am", "is" or "be" is the main verb followed by an adjective then "be" should be the predicate and the adjective the object,
for example "wind is cold" should become {"subject": "wind", "predicate": "be", "object": "cold"}.
If "am", "is" or "be" is the main verb followed by a noun phrase then "be" should be the predicate and the noun phrase the object,
for example "this is a teddy bear" should become {"subject": "this", "predicate": "be", "object": "teddy-bear"}.
Do not extend the predicate with hyphens followed by determiners such as "a", "an", "the" or adjectives such as "cold", "big".
If the predicate is followed by an infinite verb phrase with "to" such as "like to swim",
then the object should start with "to-" as in "to-swim".
If the predicate is an cognitive verb such as "think", "believe" or "know" or a speech-act such "say" or "tell",
then extract the triple from the complement phrase of the predicate,
for example "I think selene likes cats" should become {"subject": "selene", "predicate": "like", "object": "cats"}.
Combine multi-word subjects, predicates and objects with hyphens, for example "three white cats" should become "three-white-cats".
If you have no value for the subject or object use an empty string "" as a value, do NOT use none or None as a value.
Ensure that predicates are semantically meaningful.

Additionally, use any auxiliary verbs, negation words, adverbs and cognitive verbs to annotate each triple with:
    - Sentiment (-1.0 for negative, 0 for neutral, 1.0 for positive)
    - Polarity (-1.0 for negation and 1.0 for affirmation, no other values)
    - Certainty (a scale between 0 for uncertain and 1.0 for certain)
```

```
Only use floats as values for Sentiment, Polarity and Certainty. Do NOT use 0.0 as a value for Polarity.

Save it as a JSON with this format:
{"sender": "user", "text": "I am from Amsterdam.", "triples": [ { "subject": "I", "predicate": "be-from", "object": "Amsterdam",
"perspective": {"sentiment": 0, "polarity": 1, "certainty": 1}}]},
{"sender": "user", "text": "lana is reading a book.", "triples": [ { "subject": "lana", "predicate": "read", "object": "book",
"perspective": {"sentiment": 0, "polarity": 1, "certainty": 1}}]},
{"sender": "user", "text": "You hate dogs.", "triples": [ { "subject": "You", "predicate": "hate", "object": "dogs",
"perspective": {"sentiment": -1, "polarity": 1, "certainty": 0.7}}]},
{"sender": "user", "text": "Selene does not like cheese.", "triples": [ { "subject": "selene", "predicate": "like", "object": "cheese",
"perspective": {"sentiment": -1, "polarity": -1, "certainty": 0.5}}]},
{"sender": "user","text": "Selene likes to swim", "triples": [ {"subject": "selene", "predicate": "like", "object": "to-swim",
"perspective": {"sentiment": 1, "polarity": 1, "certainty": 0.1}}]}
{"sender": "user","text": "Selene likes swimming", "triples": [ {"subject": "selene", "predicate": "like", "object": "swimming",
"perspective": {"sentiment": 1, "polarity": 1, "certainty": 0.1}}]}
{"sender": "user","text": "I have to go to paris", "triples": [ {"subject": "i", "predicate": "go-to", "object": "paris",
"perspective": {"sentiment": 1, "polarity": 1, "certainty": 0.1}}]}
{"sender": "user","text": "I think that you like cats", "triples": [ {"subject": "you", "predicate": "like", "object": "cats",
"perspective": {"sentiment": 1, "polarity": 1, "certainty": 0.1}}]}
{"sender": "user","text": "John said you like cats", "triples": [ {"subject": "you", "predicate": "like", "object": "cats",
"perspective": {"sentiment": 1, "polarity": 1, "certainty": 0.1}}]}
Do not output any other text than the JSON.'''
}

class QUESTION:
    INSTRUCT = {'role':'system', 'content': '''You will receive a question from a user in a conversation
    and you need to break it down into triples consisting of a subject, predicate,and object.
If the question is a yes/no question that starts with a verb,
the triple should capture the essence of question by the speaker,
for example "Do cats eat grass?" should be represented as {"subject": "cats", "predicate": "eat", "object": "grass"}.
If the question contains a wh-word such as "who", whom", "what", "where", or "when"
then represent the triple element for the wh-word as an empty string "",
for example "Who eats grass?" should be represented as {"subject": "", "predicate": "eat", "object": "grass"},
"What do you eat? should be represented as  {"subject": "you", "predicate": "eat", "object": ""}.
If the wh-word is preceded or followed by a preposition such as "by whom", "for what", "to where" or "who for", "what for", "where to",
extend the predicate with the a hyphen followed by the preposition,
for example "To where did Selene go?" should become {"subject": "selene", "predicate": "go-to", "object": ""}.
If the question contains "how" give an empty string for the object element but add "-by" to the predicate,
for example "How did Selene come?" should become  {"subject": "selene", "predicate": "come-by", "object": ""}.
If the question contains "why" give an empty string for the object element but add "-because" to the predicate,
for example "Why did Selene come?" should become  {"subject": "selene", "predicate": "come-because", "object": ""}.
Replace the predicate by its lemma, for example "is" and "am" should become "be", "likes" and "liked" should become "like".
Remove auxiliary verbs from the predicates such as "be", "have", "can", "might", "must", "will", "shall", "should",
and also negation variants such as "cannot", "won't", "shouldn't".
Remove negation words such as "n't", "no", "not" and "never" from the predicates.
If the object starts with a preposition, concatenate the preposition to the predicate separated by a hyphen,
for example "Who is from Mexico?" should become {"subject": "", "predicate": "be-from", "object": "mexico"}.
If "am", "is" or "be" is the main verb followed by an adjective then "be" should be the predicate and the adjective the object,
for example "what is cold" should become {"subject": "", "predicate": "be", "object": "cold"}.
If "am", "is" or "be" is the main verb followed by a noun phrase then "be" should be the predicate and the noun phrase the object,
for example "what is a teddy bear" should become {"subject": "teddy-bear", "predicate": "be", "object": ""}.
Do not extend the predicate with determiners such as "a", "an", "the" or adjectives such as "cold", "nice".
If the predicate is followed by an infinite verb phrase with "to" such as "like to swim",
then the object should start with "to-" as in "to-swim".
Combine multi-word subjects, predicates and objects with an hyphens, for example 'my-sister", "three-white-cats".
Ensure that predicates are semantically meaningful.
Save it as a JSON with this format:
{"sender": "user","text": " Who likes cats?", "triples": [ {"subject": "", "predicate": "like", "object": "cats"}]},
{"sender": "user","text": " When did Selene come?", "triples": [ {"subject": "Selene", "predicate": "come", "object": ""}]},
{"sender": "user","text": " Where can I go?", "triples": [ {"subject": "I", "predicate": "go", "object": ""}]},
{"sender": "user","text": " Who likes cats?", "triples": [ {"subject": "", "predicate": "like", "object": "cats"}]},
{"sender": "user","text": " What are cats?", "triples": [ {"subject": "cats", "predicate": "be", "object": ""}]},
{"sender": "user","text": " Are cats pets?", "triples": [ {"subject": "cats", "predicate": "be", "object": "pets"}]},
{"sender": "user","text": " Do you like cats?", "triples": [ {"subject": "you", "predicate": "like", "object": "cats"}]}
Do not output any other text than the JSON.'''
}
```

Table 6. Predicates detected by the conversational triple extractor

act	become	cut	find	know	occupy	recycle	socialize	waste
add	believe	dance	finish	lead	open	relationship	sort	watch
afford	betray	decide	fit	learn	order	relax	speak	water
agree	better than	depend on	fix	leave	owe	remain	spend	wear
aid	borrow	destroy	follow	light	own	remember	start	will
allow	break	devote	forget	like	paint	remove	starve	win
anticipate	break up	die	gain	limit	park	reserve	stay	work
arrest	bring	different	get	listen	pay	respond	steal	work with
arrive	burn	dislike	get in	live	pay attention	retire	stop	worry
ask	buy	distrust	give	live in	pick	return	study	would
be	call	do	go	located	plan	rule	survive	write
be angry	can	do badly	grow	lock out	play	run	swim	
be arrested	cancel	do well	handle	look	portray	save	take	
be at	care	draw	harder than	lose	practice	say	take off	
be available	catch	dress	have	love	pray	see	take out	
be certain	cause	drink	hear	made of	prefer	seek	talk	
be difficult	change	drive	help	make	prepare	seem	taste	
be free	check in	drop	hit	marry	pretend	sell	teach	
be from	choose	earn	hold	match	prevent	send	tell	
be happy	clean	eat	hope	mean	promise	share	think	
be in	close	enable	hurry	meet	propose	shoot	throw	
be in between	come	endure	hurt	miss	protect	should	tired	
be old	confuse	equal	include	misspell	pull	show	travel	
be on	consider	exchange	investigate	mix	put	sign	try	
be out	contact	exercise	involve	more expensive	rain	similar	use	
be over	cook	expect	join	motivate	raise	sing	visit	
be ready	copy	fall	joke	move	reach	sit	wait	
be scared	cost	feel	keep	must	read	sleep	wake up	
be with	could	fight	keep secret	need	recognize	smell	walk	
be wrong	count	fill	kill	None	recommend	smoke	want	

Table 7. Breakdown of mismatches in perspectives values for the different models. The value in the Mismatch column is the gold value. > and < indicate that the predicted value is higher or lower. For polarity, we give both the predicted and gold value in this order.

Perspective	Mismatch	CFG	convALBERT	convBERT	Llama3.2	Qwen2.5
certainty	0.75	7		4		6
	< 0.75				6	
	< 1.0				10	10
polarity	1.0 != -1.0	4	6	9	2	
	0.0 != -1.0				3	
	0.5 != 1.0				1	
sentiment	+ -1	5	2	5	1	2
	> 0				8	6
	> 0.75	4	2	4	4	4
	< 1	2	2	1		